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# ODD END SEMESTER EXAMINATION DECEMBER – 2025

Name of the Course: M. Tech Structural Engineering Semester: III
Name of the Paper: Soil-Structure Interaction
Paper Code: MSE315
Time: 3 Hours Maximum Marks: 100

**Note:-**

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

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| **Q1** | **(20marks)** | |
| (a) | A building with a fundamental fixed-base period of 0.8 seconds is constructed on soil with an average effective shear wave velocity of 200 m/s. Estimate the fundamental period of the structure considering SSI effects using an appropriate empirical relation, and comment on the influence of SSI on structural response. | |
| (b) | Identify and explain situations where SSI effects are most significant, such as: Large building footprints; Substantial foundation embedment; High structure-to-soil stiffness ratios; Foundation rocking impacts. Provide examples illustrating each condition. | **CO1** |
| (c) | Discuss the different types of Soil-Structure Interaction modeling approaches — such as direct, substructure, and simplified methods. Highlight the advantages and limitations of each. | |
| **Q2** | **(20 marks)** | |
| (a) | What are the limitations and challenges of incorporating embedment effects in SSI models? Illustrate your answer with examples showing when embedment effects significantly alter structural response predictions. | |
| (b) | How is the Depth of Embedment determined for different types of foundations? Explain the criteria and parameters considered in estimating embedment depth with relevant examples. | **CO2** |
| (c) | Discuss the major limitations and issues associated with Base Slab Averaging when used in practical SSI analysis. Provide examples where this method might lead to inaccurate results. | |
| **Q3** | **(20 marks)** | |
| (a) | What is meant by bounding analyses in the context of foundation flexibility? How do bounding methods help in defining upper and lower limits of soil–foundation stiffness? | **CO3** |
| (b) | Compare and contrast the following three methods for modeling vertical and rotational foundation flexibility:<br>Method 1: Rigid Foundation on Flexible Soil<br>Method 2: Flexible Foundation on Nonlinear Flexible Soil<br>Method 3: Flexible Foundation on Linear Flexible Soil<br>Include the key assumptions, governing equations, and typical applications of each. | |
| (c) | What soil properties are essential for flexibility calculations? Describe how these properties influence the determination of vertical and rotational spring constants for a foundation. | |
| **Q4** | **(20 marks)** | |
| (a) | Derive and explain the key equations used to calculate the natural period of a | |

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| | structure for both the fixed-base and flexible-base conditions. Discuss the parameters influencing the change in period. | CO4 |
| (b) | Illustrate the dynamic response of a structure with and without soil flexibility. Discuss how period lengthening influences base shear and inter-story drift demands. | |
| (c) | Discuss the limitations and practical issues associated with estimating foundation damping and period lengthening effects in design. How do codes of practice simplify these aspects for engineering use? | |
| **Q5** | **(20 marks)** | CO5 |
| (a) | Explain the concept of basement modelling in the context of soil–structure interaction (SSI). Discuss the importance of accurately representing basements in seismic analysis and design of buildings. | |
| (b) | Describe the types of basement models that have been studied in past research and identify the recommended models for practical design applications. Justify why certain models are preferred over others. | |
| (c) | Explain how basement modelling for buildings on sloping grade differs from those on level grade. Provide an example highlighting the influence of slope geometry on lateral earth pressure and structural response. | |